Infinity Learn

OLYMPIAD WORKSHEET

Time and Motion

Science — Physics · Class 7 · Max Marks: 17 · Time: 45 min

Section A — Multiple Choice (Higher Order) (1 × 14 = 14)

1. A cyclist covers 3 km in 10 minutes. His speed is:

A. 0.3 km/h
B. 18 km/h
C. 30 km/h
D. 3 km/h

2. A car moves at 15 m/s. The distance it covers in 2 minutes is:

A. 1800 m
B. 900 m
C. 450 m
D. 30 m

3. A speed of 54 km/h is equal to:

A. 150 m/s
B. 5.4 m/s
C. 54 m/s
D. 15 m/s

4. Two stations are 300 km apart. A train leaves at 8:00 a.m. and arrives at 12:00 noon. Its average speed is:

A. 300 km/h
B. 25 km/h
C. 75 km/h
D. 60 km/h

5. On a distance–time graph, the line of body P is steeper than that of body Q. This means:

A. P is faster than Q
B. P is slower than Q
C. both have the same speed
D. P is at rest

6. A pendulum takes 40 s to complete 20 oscillations. Its time period is:

A. 0.5 s
B. 2 s
C. 40 s
D. 20 s

7. A bus travels the first 60 km at 60 km/h and the next 60 km at 30 km/h. Its average speed for the whole journey is:

A. 30 km/h
B. 40 km/h
C. 45 km/h
D. 90 km/h

8. If the length of a pendulum is increased, its time period will:

A. increase
B. stay the same
C. become zero
D. decrease

9. A body that stays at rest for 5 seconds is shown on a distance–time graph as:

A. a slanting straight line
B. a horizontal line
C. a vertical line
D. a curve rising steeply

10. A car's odometer reads 5642 km at the start of a trip and 5792 km at the end, 3 hours later. Its average speed was:

A. 1934 km/h
B. 45 km/h
C. 50 km/h
D. 150 km/h

11. Which of the following is the fastest?

A. 12 m/s
B. 600 m/min
C. 36 km/h
D. 1 km/min

12. A runner completes one lap of a 400 m circular track in 50 s. Her speed is:

A. 20 m/s
B. 400 m/s
C. 8 m/s
D. 50 m/s

13. A pendulum clock is running slow. To make it keep correct time, the length of its pendulum should be:

A. decreased
B. doubled
C. increased
D. kept unchanged

14. Two buses start towards each other from towns 240 km apart, at 40 km/h and 60 km/h. They meet after:

A. 3 hours
B. 4 hours
C. 6 hours
D. 2.4 hours

Section B — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** On a distance–time graph, a horizontal line means the body is at rest.
   Reason (R): For a body at rest the distance does not change with time.
2. **Assertion (A):** A speed of 72 km/h is the same as 20 m/s.
   Reason (R): To convert km/h into m/s we multiply by 5/18.
3. **Assertion (A):** A simple pendulum is used in clocks to measure time.
   Reason (R): Each oscillation of a given pendulum takes a different amount of time.

Answer Key

Multiple Choice (Higher Order)

1. B	**2.** A	**3.** D	**4.** C	**5.** A
6. B	**7.** B	**8.** A	**9.** B	**10.** C
11. D	**12.** C	**13.** A	**14.** D	

Assertion–Reason

1. A
2. A
3. C